1 Proof of Interval of Convergence Theorem

We will now prove the theorem from last time:

Theorem 1.1. *For a given power series $\sum_{k=0}^{\infty} c_k(x-a)^k$ there are only three possibilities:*

(i) The series converges only when $x = a$.

(ii) The series converges for all x.

(iii) There is a positive number R such that the series converges if $|x-a| < R$ and diverges if $|x-a| > R$.

It suffices to prove Theorem 1.1 for the case $a = 0$ (if $a \neq 0$, we can set $u = x - a$ and prove it for the variable u). We will need the following lemma:

Lemma 1.2. *(a) If a power series $\sum_{k=0}^{\infty} c_k x^k$ converges absolutely at $x = b \neq 0$, then it converges whenever $|x| < |b|$.*

(b) If a power series $\sum_{k=0}^{\infty} c_k x^k$ diverges at $x = d \neq 0$, then it diverges whenever $|x| > |d|$.

Proof. (of Lemma 1.2)

1. We first prove part (a). If $\sum_{k=0}^{\infty} c_k b^k$ converges, then $\lim_{k\to\infty} c_k b^k = 0$.

   It follows that there exists a positive integer N such that $|c_k b^k| < 1$ whenever $n \geq N$. Thus, for $n \geq N$ and any x, we have

   $$0 < |c_k x^k| = \left|\frac{c_k b^k x^k}{b^k}\right| = |c_k b^k| \left|\frac{x}{b}\right|^k < \left|\frac{x}{b}\right|^k. \tag{1.3}$$

2. Use (1.3) to conclude that if $|x| < |b|$, then $\sum c_k x^k$ converges absolutely, and hence converges (*Hint*: Use the Comparison Test.)
   If $|x| < |b|$, then $\left|\frac{x}{b}\right| < 1$, so $\sum_k |\frac{x}{b}|^k$ is a convergent geometric series. Since $0 < |c_k x^k| < \left|\frac{x}{b}\right|^k$, by the Comparison Test $\sum |c_k x^k|$ converges. This shows that $\sum c_k x^k$ converges absolutely, and hence converges.

3. Suppose $\sum c_k d^k$ diverges. Let x be any number such that $|x| > |d|$. If $\sum c_k x^k$ converges, then part (a) leads to a contradiction. What is the contradiction?
   If $|x| > |d|$ and $\sum c_k x^k$ converges, then by part (a) $\sum c_k y^k$ converges for all $|y| < |x|$, and in particular for $y = d$. This contradicts the assumption that $\sum c_k d^k$ diverges.

□

Let

$$R = \sup\{|x| : \sum_{k=0}^{\infty} a_k x^k \text{converges}\}$$

4. Prove that if $R = 0$, then the series converges only for $x = 0$.
   Suppose $\sum_{k=0}^{\infty} a_k x^k$ converges for some x_0. Then $0 \leq |x_0| \leq R = 0$ implies $|x_0| = 0$ and therefore $x_0 = 0$.

5. Prove that if $R > 0$, then the series converges absolutely if $|x| < R$ and diverges for $|x| > R$.
   If $|x| < R$, then $|x|$ is not an upper bound for the set $\{|x| : \sum_{k=0}^{\infty} a_k x^k \text{converges}\}$. Thus, there exists x_0 such that $\sum_{k=0}^{\infty} a_k x_0^k$ converges and $|x| < |x_0|$. By Lemma 1.2, $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely. If $|x| > R$, then the series diverges by definition of R (since if the series converged at x, then $|x| \leq R$, which is a contradiction).

6. Prove that if $R = \infty$, then the series converges for all $x \in \mathbb{R}$.
   If $R = \infty$, then the set $\{|x| : \sum_{k=0}^{\infty} a_k x^k \text{converges}\}$ is not bounded above, hence the series converges for all real numbers.

Note that Theorem 1.1 does not say what happens at the endpoints $x = a \pm R$, and in general the power series may converge or diverge there. We refer to the set of all points where the power series converges as its interval of convergence, which is one of

$$(a-R, a+R), \quad (a-R, a+R], \quad [a-R, a+R), \quad [a-R, a+R].$$

Theorem 1.1 also does not give an explicit expression for the radius of convergence of a power series in terms of its coefficients. The ratio test gives a simple, but useful, way to compute the radius of convergence, although it doesn't apply to every power series.

Theorem 1.4. *Suppose $a_n \neq 0$ for infinitely many n and the limit*

$$R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right|$$

exists or diverges to infinity. Then the power series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

has radius of convergence R.

Proof. Let

$$r = \lim_{n\to\infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n (x-a)^n} \right| = |x-a| \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| = \frac{|x-a|}{R}.$$

By the Ratio Test, the power series converges if $0 \leq r < 1$, or $|x-a| < R$, and diverges if $1 < r \leq \infty$, or $|x-a| > R$, which proves the result. □

The root test also gives a simple way to compute the radius of convergence. This method can be extended to *all* power series by introducing the *lim sup* of a sequence (which always exists), but we won't need this level of generality in this course.

Theorem 1.5. *The radius of convergence R of the power series*

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

is given by

$$R = \frac{1}{\lim_{n\to\infty} |a_n|^{1/n}},$$

provided the limit exists or diverges to infinity. [1]

Proof. Let $r = \lim_{n\to\infty} |c_n (x-a)^n|^{1/n} = |x-a| \lim_{n\to\infty} |a_n|^{1/n} = \frac{|x-a|}{R}$. By the Root Test, the series converges when $0 \leq r < 1$, or $|x-a| < R$, and diverges if $1 < r \leq \infty$, or $|x-a| > R$, which proves when the result. □

[1] More generally, for *any* power series the radius of convergence is given by $R = \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}$.

2 Uniform Convergence

Let $f_n(x) = c_n x^n$. Then convergence of $f(x) = \sum_{n=0}^{\infty} c_n x^n$ on $(-R,R)$ means that, given $\varepsilon > 0$, for all $x \in (-R,R)$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|f(x) - f_n(x)| < \varepsilon$. One says that the sequence of functions $\{f_n\}$ *converges pointwise* to the function f. However, something stronger turns out to be true. It turns out to be the case that, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $x \in (-R,R)$, $n \geq N$ implies $|f_n(x) - f(x)| < \varepsilon$. The quantifier structure in the first statement is $\forall\varepsilon\forall x\exists N$, while in the latter it is $\forall\varepsilon\exists N\forall x$; that is, the same N works for all $x \in (-R,R)$. In this case, the sequence of functions $\{f_n\}$ is said to *converge uniformly* on $(-R,R)$ to the function f.

Uniform convergence has important consequences for the function f, which are given in the three theorems below.

Theorem 2.1. *Suppose that $\{f_n\}$ is a sequence of functions which are continuous on $[a,b]$, and that $\{f_n\}$ converges uniformly on $[a,b]$ to f. Then f is continuous on $(-R,R)$.*

Proof. We will consider the case $x \in (a,b)$. The proof for $x = a$ or $x = b$ requires the usual modifications. Let $\varepsilon > 0$. Since $\{f_n\}$ converges uniformly to f on $(-R,R)$, there exists n such that

$$|f(y) - f_n(y)| < \frac{\varepsilon}{3} \text{ for all } y \in (-R,R).$$

In particular, for all h such that $x+h \in (-R,R)$, we have

$$|f(x) - f_n(x)| < \frac{\varepsilon}{3},$$
$$|f(x+h) - f_n(x+h)| < \frac{\varepsilon}{3}.$$

Since f_n is continuous, there exists $\delta > 0$ such that for $|h| < \delta$ we have

$$|f_n(x) - f_n(x+h)| < \frac{\varepsilon}{3}.$$

Thus, if $|h| < \delta$ then

$$\begin{aligned} |f(x+h) - f(x)| &= |f(x+h) - f_n(x+h) + f_n(x+h) - f_n(x) + f_n(x) - f(x)| \\ &\leq |f(x+h) - f_n(x+h)| + |f_n(x+h) - f_n(x)| + |f_n(x) - f(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon, \end{aligned}$$

which proves that f is continuous on $(-R,R)$. □

Theorem 2.2. *Suppose that $\{f_n\}$ is a sequence of functions which are integrable on $[a,b]$, and that f_n converges uniformly on $[a,b]$ to a function f which is integrable on $[a,b]$. Then*

$$\int_a^b f(x)dx = \lim_{n\to\infty} \int_a^b f_n(x)dx.$$

Since $f(x) = \lim_{n\to\infty} f_n(x)$, this says that

$$\int_a^b \lim_{n\to\infty} f_n(x)dx = \lim_{n\to\infty} \int_a^b f_n(x)dx.$$

Proof. Let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have

$$|f(x) - f_n(x)| < \frac{\varepsilon}{b-a} \text{ for all } x \in [a,b].$$

Thus, if $n \geq N$, we have

$$\begin{aligned}\left|\int_a^b f(x)dx - \int_a^b f_n(x)dx\right| &= \left|\int_a^b [f(x) - f_n(x)]dx\right| \\ &\leq \int_a^b |f(x) - f_n(x)|dx \\ &< \int_a^b \frac{\varepsilon}{b-a}dx \\ &= \frac{\varepsilon}{b-a}\int_a^b dx \\ &= \frac{\varepsilon}{b-a}(b-a) \\ &= \varepsilon,\end{aligned}$$

which proves that $\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b f(x)dx$. □

Unfortunately, if each f_n is differentiable and $\{f_n\}$ converges uniformly to f, it is not necessarily true that f is differentiable. However, the Fundamental Theorem of Calculus practically guarantees that some sort of theorem about derivatives will be a consequence of Theorem 2.2; the crucial hypothesis is that $\{f_n'\}$ converges uniformly to *some* continuous function.

Theorem 2.3. *Suppose $\{f_n\}$ is a sequence of functions which are differentiable on $[a,b]$ with integrable derivatives f_n', and that $\{f_n\}$ converges pointwise to f. Suppose, moreover, that $\{f_n'\}$ converges uniformly on $[a,b]$ to some continuous function g. Then f is differentiable and*

$$f'(x) = \lim_{n\to\infty} f_n'(x).$$

Proof. Applying Theorem 2.2 to the interval $[a,x]$, we see that for each x we have

$$\begin{aligned}\int_a^x g(x)dx &= \lim_{n\to\infty}\int_a^x f_n'(x)dx \\ &= \lim_{n\to\infty}[f_n(x) - f_n(a)] \\ &= f(x) - f(a).\end{aligned}$$

Since g is continuous, it follows that $f'(x) = g(x) = \lim_{n\to\infty} f_n'(x)dx$ for all $x \in [a,b]$. □

We can now apply each of Theorems 2.1, 2.2, and 2.3 to uniformly convergent series; the results may be stated in one common corollary.

Corollary 2.4. *Suppose $\sum_{n=0}^{\infty} f_n$ converges uniformly to f on $[a,b]$.*

(1) If each f_n is continuous on $[a,b]$, then f is continuous on $[a,b]$.

(2) If f and each f_n is integrable on $[a,b]$, then

$$\int_a^b f(x)dx = \sum_{n=0}^{\infty} \int_a^b f_n(x)dx$$

Moreover, if $\sum_{n=0}^{\infty} f_n$ converges pointwise to f on $[a,b]$, each f_n has an integrable derivative f_n' and $\sum_{n=0}^{\infty} f_n'$ converges uniformly on $[a,b]$ to some continuous function, then

(3) $f'(x) = \sum_{n=1}^{\infty} f_n'(x)$ for all $x \in [a,b]$.

Proof. (1) If each f_n is continuous, then so is each $s_n := \sum_{k=1}^{n} f_n$. Since $\{s_n\}$ converges uniformly to f, f is continuous by Theorem 2.1.

(2) Since $\{s_n\}$ converges uniformly to f, it follows from Theorem 2.2 that

$$\begin{aligned}\int_a^b f(x)dx &= \lim_{n\to\infty} \int_a^b \sum_{k=0}^{n} f_n(x)dx \\ &= \lim_{n\to\infty} \sum_{k=0}^{n} \int_a^b f_n(x)dx.\end{aligned}$$

(3) Each $s_n(x) = \sum_{k=0}^{n} f_n(x)$ is differentiable with $s_n'(x) = \sum_{k=0}^{n} f_n'(x)$. Since $\{s_n'\}$ converges uniformly to a continuous function, it follows from Theorem 2.3 that

$$\begin{aligned}f'(x) &= \lim_{n\to\infty} s_n'(x) \\ &= \sum_{n=0}^{\infty} f_n'(x).\end{aligned}$$

□

To prove that a convergent power series converges uniformly on its interval of convergence, we will use the following test.

Theorem 2.5 (Weierstrass M-test)**.** *Let $\{f_n\}$ be a sequence of functions defined on A, and suppose that $\{M_n\}$ is a sequence of numbers such that*

$$|f_n(x)| \le M_n \text{ for all } x \in A.$$

Suppose moreover that $\sum_{n=0}^{\infty} M_n$ converges. Then for each $x \in A$ the series $\sum_{n=0}^{\infty} f_n(x)$ converges absolutely, and $\sum_{n=1}^{\infty} f_n$ converges uniformly on A to the function

$$f(x) = \sum_{n=1}^{\infty} f_n(x).$$

For each $x \in A$ the series $\sum_{n=0}^{\infty} |f_n(x)|$ converges by the Comparison Test, hence $\sum_{n=0}^{\infty} f_n(x)$ converges absolutely. Moreover, given $\varepsilon > 0$, for each $x \in A$ we have

$$\begin{aligned}\left| f(x) - \sum_{k=0}^{N} f_N(x) \right| &= \left| \sum_{k=N+1}^{\infty} f_N(x) \right| \\ &\leq \sum_{k=N+1}^{\infty} |f_N(x)| \\ &\leq \sum_{k=N+1}^{\infty} M_N.\end{aligned}$$

Since $\sum_{n=0}^{\infty} M_n$ converges, $\lim_{n\to\infty} M_n = 0$, so the number $\sum_{k=N+1}^{\infty} M_n$ can be made less than ε by choosing N sufficiently large.

We are now ready to prove that a power series converges uniformly and absolutely on its interval of convergence.

Theorem 2.6. *Suppose the series*

$$f(x_0) = \sum_{n=0}^{\infty} c_n x_0^n$$

converges, and let a be any number with $0 < a < |x_0|$. Then on $[-a, a]$ the series

$$f(x) = \sum_{n=0}^{\infty} c_n x^n$$

converges uniformly and absolutely. Moreovoer, the same is true for the series

$$g(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}.$$

Finally, f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

for all x with $|x| < |x_0|$.

Proof. Since $\sum_{n=0}^{\infty} c_n x_0^n$ converges, $\lim_{n\to\infty} c_n x_0^n = 0$. Hence they are bounded: there is some positive number M such that

$$|c_n x_0^n| = |c_n|\ |x_0^n| \leq M \text{ for all } n.$$

Now if $x \in [-a, a]$, then $|x| \leq |a|$, so

$$\begin{aligned}|c_n x^n| &= |c_n|\ |x^n| \\ &\leq |c_n|\ |a^n| \\ &= |c_n|\ |x_0|^n \left| \frac{a}{x_0} \right|^n \\ &\leq M \left| \frac{a}{x_0} \right|^n.\end{aligned}$$

Since $\frac{a}{x_0} < 1$, the geometric series

$$\sum_{n=0}^{\infty} M\left|\frac{a}{x_0}\right|^n = M\sum_{n=0}^{\infty}\left|\frac{a}{x_0}\right|^n$$

converges. Choosing $M\left|\frac{a}{x_0}\right|^n$ as the number M_n in the Weierstrass M-test, it follows that $\sum_{n=0}^{\infty} c_n x^n$ converges uniformly on $[-a,a]$.

To prove the same assertion for $g(x) = \sum_{n=1}^{\infty} nc_n x^{n-1}$ notice that

$$\begin{aligned}|na_n x^{n-1}| &= n|c_n|\ |x^{n-1}| \\ &\le n|c_n|\ |a^{n-1}| \\ &= \frac{|c_n|}{|a|}|x_0|^n\, n\left|\frac{a}{x_0}\right|^n \\ &\le \frac{M}{|a|}n\left|\frac{a}{x_0}\right|^n.\end{aligned}$$

Since $\left|\frac{a}{x_0}\right| < 1$, the series

$$\sum_{n=1}^{\infty}\frac{M}{|a|}n\left|\frac{a}{x_0}\right|^n = \frac{M}{|a|}\sum_{n=1}^{\infty} n\left|\frac{a}{x_0}\right|^n$$

converges by the Ratio Test. Another appeal to the Weierstrass M-test proves that $\sum_{n=1}^{\infty} na_n x^{n-1}$ converges uniformly on $[-a,a]$.

Finally, Corollary 2.4 proves, first that g is continuous, and then that

$$f'(x) = g(x) = \sum_{n=1}^{\infty} nc_n x^{n-1} \text{ for } x \in [-a,a].$$

Since we could have chosen any number a with $0 < a < |x_0|$, this result holds for all x with $|x| < |x_0|$. Since x_0 was an arbitrary point at which the power series converges, this proves that f' has the same interval of convergence as f.

Since f is continuous on its interval of convergence, it is integrable. Since each f_n is integrable, by part (2) of Corollary 2.4, for any interval $[a,x]$ contained in the interval of convergence, we have

$$\int_a^x f(t)dt = \int_a^x \sum_{n=0}^{\infty} c_n t^n dt = \sum_{n=0}^{\infty}\int_a^x c_n t^n dt = \sum_{n=0}^{\infty} c_n \frac{x^{n+1} - a^{n+1}}{n+1}.$$

Since x can be any value in the interval of convergence of f, the integrated power series on the right has the same interval of convergence as f. □